Load Controlled Tensile Testing of Copper Wire

Addeo, Bryce
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***Abstract*—This experiment utilized load-controlled tensile testing to analyze the stress-strain relationship of annealed 32 American Wire Gauge (AWG) copper wire. The top of the wire was secured to a loading apparatus, and a hanger was suspended from the bottom. The hanger featured a metal core that displaced into a linear variable differential transducer (LVDT). Brass slotted weights were incrementally loaded on the hanger, leading the moving core to displace inside the LVDT. This generated a differential voltage that was sent to a LabVIEW program via a SADI 2.7 Data Acquisition device (DAQ). This voltage was converted to the elongation of the wire with a calibration constant which allowed for the calculation of strain. The stress was calculated through the incremental weight and wire cross-sectional area inputs. A short wire was first tested until it reached a breaking point to determine the ultimate tensile strength, and a long wire was then tested to analyze the elastic region and yield stress of the copper wire.**

The stress and strain data for both wires were plotted on a graph using excel. An analysis of this data generated a modulus of elasticity equal to 66.16 ± 17.94 GPa, a yield stress of 102.16 MPa, an offset yield point stress of 110.49 MPa, and an ultimate tensile strength of 229.72 ± 0.573 MPa. A comparison of these experimental results to the known material properties of the copper wire revealed a considerable difference in the values. This disparity was produced from experimental limitations such as wire slip, creep, and human measurement errors, as well as the multiple sources of uncertainty including given, statistical, RSS propagated, and Monte Carlo simulated uncertainties. Proposed improvements involved a better clamping technique to reduce slip, more precise initial wire length measurements, and temperature control to mitigate creep.

***Index Terms*—Linear Variable Differential Transducer, Data Acquisition Device, Load Controlled Tensile Testing, Stress-Strain Curve**

I. Introduction

THE main objective of this lab was to perform load controlled tensile testing on a copper wire to characterize the elastic and plastic regions of its response [6]. To achieve this objective, a SADI 2.7 Data Acquisition device (DAQ) was connected to a linear variable differential transducer (LVDT) on an apparatus that allowed for the incremental loading of a copper wire [6]. A LabVIEW program recorded the voltage associated with the change in length of the copper wire and calculated stress and strain based off that data and other inputs. Limitations in measuring techniques and error from multiple sources led to uncertainty in the results.

Load controlled tensile testing involves imposing uniaxial stress to induce a resultant strain [6]. In this lab, weights were placed on a hanger connected to a copper wire to generate downward stress through its cross-sectional area. The stress was applied in tension since the weights pulled down and stretched the wire. Compressive forces were not applied in this experiment as that would have squeezed and pushed the material inward. Since tensile loading was utilized, the wire experienced a change in length in the same axial direction that the load was applied. Equations for stress and strain in the axial direction, x, are shown below (1) and (2).

$$\sigma_x = \frac{P}{A} \tag{1}$$

$$\varepsilon_x = \frac{\Delta L}{L} \tag{2}$$

For the stress equation (1), σ_x is stress in the axial direction, P is the internal force, and A is the cross-sectional area at the location of the force [3]. For the strain equation (2), ε_x is the strain in the axial direction, ΔL is the change in length or elongation and L is the original length [3]. Although it was found from an online source for this lab, the calculation for cross-sectional area of the wire, A, is shown below in (3), where d is its diameter.

$$A = \frac{\pi}{4} d^2 \tag{3}$$

For the case of this lab, A and L represented the original cross-sectional area of the copper wire and the original length of the copper wire being tested, respectively. The cross-sectional area of the wire decreases while undergoing tensile deformation in a phenomenon known as necking. Despite this, the original area was still used in all calculations, thus generating results for engineering stress and strain rather than true stress and strain which would have considered the change in area from tensile deformation [3].

A typical stress-strain response of a metal involves elastic and plastic deformation. The initial response to loading is a linearly elastic relationship between stress and strain with a slope defined by Young's modulus (also called modulus of elasticity), E, as shown in (4) below.

$$\sigma_x = E\varepsilon_x \tag{4}$$

If the load is removed within the elastic region, then no permanent deformation will occur to the metal. However, if the load is increased until the end of the elastic region is reached, then the metal will yield and undergo plastic deformation [3]. Many materials do not have a distinct yield point and therefore an offset approach can be used to approximate it [6]. For ductile materials, the convention is to start at a strain of 0.002 and then draw a line parallel to the elastic region until it intersects the stress strain curve. This intersection point is the offset yield point.

Once the yield point, or elastic limit, is reached, plastic deformation begins. During this stage, the strain will increase at a faster rate than experienced in the elastic region therefore leading to permanent deformation even once the load is removed. Within the plastic region, work hardening first occurs, making the metal stronger until it reaches its ultimate tensile strength [3]. Once the stress reaches this maximum, necking begins and the cross-sectional area of the metal decreases until it finally fractures. It is difficult to collect accurate data after the ultimate stress is exceeded because creep, which is dependent on temperature, causes gradual deformation even under a stable load. This stress and strain relationship along with the elastic and plastic regions were observed for the copper wire tested in this lab.

A transducer is a device that changes one physical phenomena into another. In this lab, a linear variable differential transducer (LVDT) was used to measure the elongation of the wire when various weights were loaded on the attached hanger [3]. A 14-bit Data Acquisition device (DAQ) with an analog to digital converter (ADC) capable of returning 16,383 unique values, was connected to the LVDT. The LVDT converted the displacement into a voltage that was read by the DAQ and sent to the LabVIEW program to be used in calculations for the outputs. The LVDT used in this lab produced a voltage from -5 VDC to +5 VDC for a displacement of -10 millimeters to 10 millimeters from the core [6]. A sample window corresponding to the LVDT channel in the LabVIEW program was carefully selected to contain this voltage range to avoid saturation while also maximizing resolution. The voltage generated, ΔV, and the elongation of the wire, ΔL, can be related with a calibration constant, C, and the calibration constant is found using the sensitivity of the LVDT, S, shown in (5) and (6) below.

$$\Delta L = C(\Delta V) \tag{5}$$

$$C = \frac{1}{S} \tag{6}$$

In this lab, the sensitivity of the LVDT is equal to 0.485 ± 0.0015 V/mm, and the calibration constant is therefore equal to 2.0619 ± 0.0064 mm/V by (6). This constant, along with the initial wire length, weight loaded, and cross-sectional area of the wire were input into the LabVIEW program. The program read voltage from the DAQ and then used (5) to calculate elongation. That result was used in (2) to calculate strain, and stress could already be calculated with (1) from the weight and area inputs. The weights were iteratively loaded to check for plastic deformation in the copper wire and determine the yield point. The hanger used to hold the weights applied an initial load on the copper wire, but it was not yet accounted for in the LabVIEW program.

There were multiple limitations to the load controlled tensile testing used in this lab. Force resolution was limited to the size of the provided weights [6]. Additionally, numerous students performing the experiment at once further limited the availability of various weights. Displacement range was also limited as the deflection of the longer wires would exceed the displacement range of the LVDT and therefore reduce stress on the wire before it broke. Lastly, it was difficult to measure data after the ultimate load stress was achieved due to the wire dynamically breaking after this point [6]. These limitations, in addition to the various measuring techniques used, led to a stack up of uncertainties which were analyzed to determine the precision and accuracy of results.

II. Procedure

The required materials for this lab were a DAQ, LVDT and power source, annealed 32 AWG copper wires, a hanger, slotted brass weights, a digital scale, a computer capable of running LabVIEW, a USBC cord, a loading platform apparatus, cutting pliers, a screwdriver and a digital caliper. The loading platform apparatus was already constructed with an LVDT in an adjustable mount and connected to a power supply before the lab began. Load controlled tensile tests were performed for both short and long copper wires.

The short wire, which was ideally in the range of 30 to 40 millimeters, was tested first. The pliers were used to cut a segment of copper wire off a spool. It was important to ensure this wire was longer than the final target length as it needed to be wrapped around posts on the loading platform apparatus and hanger. The wire was wrapped around posts and secured by hand tightened clamps on both the loading apparatus and hanger. A visual representation of how the wire was set up is shown in Fig. 1 below.

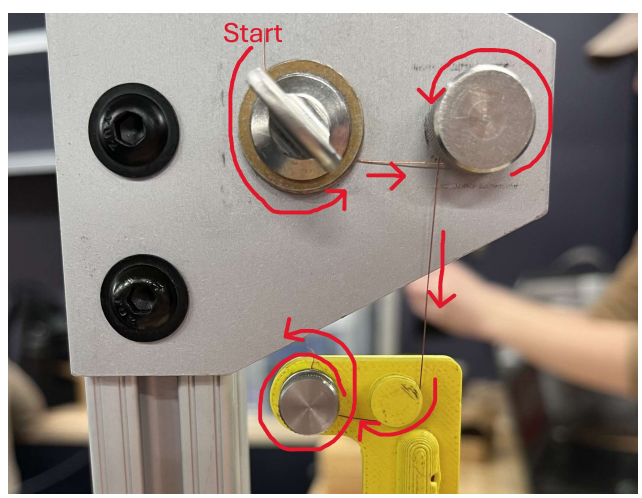

Fig. 1. Setup of wire around clamps and posts.

Once the wire was secured by the clamps, the distance between the two posts was measured with a digital caliper to approximate the initial wire length, L_{short}. The LVDT mount was then adjusted using a screwdriver so that the metal rod on the end of the hanger, called the core, was approximately in the middle of the 20-millimeter travel range in preparation for an upcoming step [3].

The 15 V power supply connected to the LVDT was turned on if not already done so. The LVDT was then connected to channel 0 on the DAQ with jumper wires, and the DAQ was connected to a laptop with the USBC cord. The LabVIEW program was loaded, and all the necessary given parameters were input. These included the previously measured initial wire length, the calibration constant equal to 2.0619 ± 0.0064, and the cross-sectional area of the copper wire according to the Engineers EDGE website equal to 0.032 ± 7.98 x 10^{-5} mm^2 [1]. The sample window for channel 0 was set equal to ±5.12 to avoid saturation and maximize resolution as this was the smallest window that contained the LVDT voltage range (±5 V).

The weights for the short wire tests were gathered and weighed in Newtons on the digital scale. Each weight was placed on the scale one at a time, without the previous ones being removed. The sum of the weights was noted for each increment added. This method of weighing the weights together helped to reduce the uncertainty of this measurement rather than measuring each weight separately and then manually adding them up. After this weighing process, the weights were brought to the apparatus and hanger setup.

Before data could be taken, the LVDT had to be correctly adjusted and tared. From the initial setup, the core attached to the hanger was already positioned approximately at the center of the 20-millimeter LVDT range. However, once the LabVIEW program was initiated a more exact reading of the differential voltage was displayed. The mount was loosened with the screwdriver and the position of the LVDT was adjusted so that the voltage reading was about equal to zero volts to get closer to the center of the range. Then, the LVDT was readjusted until a differential voltage approaching positive or negative five was achieved without exceeding either one of these values. Once this location was determined the mount was secured back in place with the screwdriver and the LVDT voltage was tared by copying and pasting its value into the tare control. The tare operation is illustrated in (7) below where ΔV_{LVDT} is the change in LVDT voltage, V_{LVDT} is the differential voltage, $V_{LVDT,tare}$ is the tare value being subtracted.

$$\Delta V_{LVDT} = V_{LVDT} - V_{LVDT,tare} \qquad (6)$$

Now that the LVDT was adjusted and tared, all inputs were in the LabVIEW program, and all weights were measured, the incremental loading process was initiated. First, one weight was added as a proof load on the hanger to remove any remaining slack in the wire and prevent slip. This proof load was removed after a few seconds and an initial data point was taken for zero weight loaded. However, an initial strain caused by the weight of the hanger was already applied to the wire although it was not yet accounted for. The first weight was placed on the hanger and its value in newtons was entered into the LabVIEW weight control. A data point was then saved, and the next weight was loaded. The sum of the two weights added up was now input into the LabVIEW weight control and a new data point was taken. This process was repeated for every weight with a data point being saved for each increment. The hanger loaded with slotted brass weights is shown in Fig. 2 below.

Fig. 2. Slotted brass weights loaded on hanger attached to copper wire.

Larger weights were first loaded for the elastic region, and then smaller weights were added for the plastic region (Fig. 2) since the response became more sensitive past the yield point. Once the load was increased to the point of the wire breaking, the program was terminated, and the data was saved to a .xls file. The broken wire was removed, and the process could then be repeated for as many trials as desired.

The data from the short wire was plotted in Microsoft Excel to produce a stress-strain graph. The approximate yield point was observed and recorded as this was useful information for the long wire trial. This was because the long wire would experience greater elongation than the LVDT range (greater than 20 mm) and would therefore not break from this testing method unlike the short wire. Hence, the goal of the long wire testing was to reach past the yield point to determine the yield strength and analyze the elastic region.

All steps for the long wire method were the same as the short wire method except for the ones detailed below. The long wire had to be greater than 60 millimeters without exceeding the length that could fit within the apparatus. This value was greater than the maximum initial wire length at which the wire would still break which was calculated using (4) and (2) with

approximate yield stress and modulus of elasticity values from the short wire data (change in length was set equal to the LVDT range of 20 mm). The wire was set up and the weights were incremented in the same fashion, however, the program was terminated once the yield stress was exceeded, not once the wire broke. Once adequate data was collected for both wire lengths, the used wires were disposed of, and the power source was turned off.

III. Results

The initial length of the short wire, L_{short}, was equal to 41.37 ± 0.5 millimeters. The measured load of each incremental weight, the associated differential LVDT voltage, and the total elongation calculated by LabVIEW using (5) are shown in Table I below for the short wire.

Table I
Load, LVDT Voltage and Elongation for Short Wire

Load (N)	LVDT Voltage (V)	Elongation (mm)
0.000 ± 0.0005	4.739 ± 0.004	0.060 ± 0.013
1.963 ± 0.0005	4.678 ± 0.005	0.185 ± 0.014
2.939 ± 0.0005	4.648 ± 0.004	0.247 ± 0.013
4.408 ± 0.0005	4.564 ± 0.004	0.420 ± 0.014
4.800 ± 0.0005	4.480 ± 0.004	0.593 ± 0.013
5.194 ± 0.0005	4.291 ± 0.005	0.983 ± 0.013
5.587 ± 0.0005	4.003 ± 0.005	1.576 ± 0.013
5.880 ± 0.0005	3.723 ± 0.004	2.154 ± 0.015
6.177 ± 0.0005	3.347 ± 0.005	2.929 ± 0.016
6.376 ± 0.0005	3.081 ± 0.005	3.478 ± 0.017
6.573 ± 0.0005	2.705 ± 0.005	4.254 ± 0.019
6.772 ± 0.0005	2.231 ± 0.005	5.230 ± 0.022
6.955 ± 0.0005	1.836± 0.009	6.045 ± 0.027
7.152 ± 0.0005	1.117 ± 0.011	7.528 ± 0.033
7.351 ± 0.0005	0.226 ± 0.024	9.366 ± 0.059

The initial length of the long wire, L_{long}, was equal to 109.92 ± 0.5 millimeters. The same parameters shown above for the short wire are shown below for the long wire in Table II.

Table II
Load, LVDT Voltage and Elongation for Long Wire

Load (N)	LVDT Voltage (V)	Elongation (mm)
0.000 ± 0.0005	4.444 ± 0.004	0.217 ± 0.011
0.980 ± 0.0005	4.420 ± 0.004	0.267 ± 0.012
1.373 ± 0.0005	4.411 ± 0.004	0.286 ± 0.012
1.760 ± 0.0005	4.401 ± 0.004	0.307 ± 0.011
2.128 ± 0.0005	4.391 ± 0.004	0.326 ± 0.012
2.517 ± 0.0005	4.379 ± 0.006	0.351 ± 0.014
2.902 ± 0.0005	4.359 ± 0.012	0.392 ± 0.026
3.269 ± 0.0005	4.342 ± 0.012	0.428 ± 0.027
3.653 ± 0.0005	4.291 ± 0.007	0.533 ± 0.017
3.850 ± 0.0005	4.262 ± 0.012	0.593 ± 0.026
4.036 ± 0.0005	4.161 ± 0.009	0.800 ± 0.020
4.220 ± 0.0005	4.015 ± 0.012	1.101 ± 0.027
4.405 ± 0.0005	3.790 ± 0.011	1.568 ± 0.025

The relationship between load and elongation from the tables above is better shown by the graph below in Fig. 3.

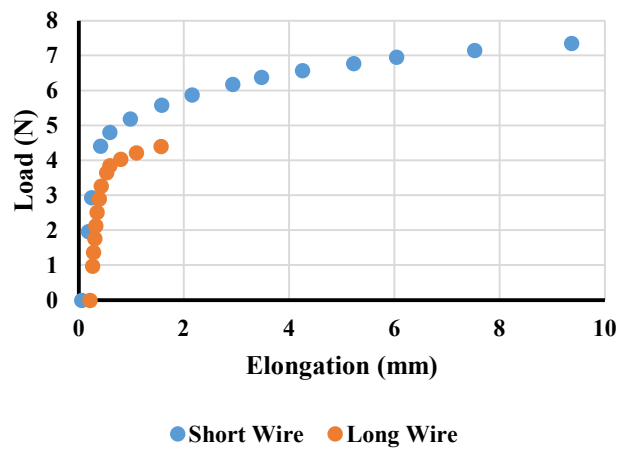

Fig. 3. Load versus elongation for short wire and long wire.

Based off the data from Tables I and II, along with the area and initial length inputs, the LabVIEW program calculated stress and strain for both wires using (2) and (3). The stress and strain results for the short wire and the long wire are presented in Tables III and IV below, respectively.

Table III
Stress and Strain for Short Wire

Strain (mm/mm)	Stress (MPa)
$1.45 \times 10^{-3} \pm 3.15 \times 10^{-4}$	0.00 ± 0.016
$4.47 \times 10^{-3} \pm 3.32 \times 10^{-4}$	61.34 ± 0.154
$5.97 \times 10^{-3} \pm 3.26 \times 10^{-4}$	91.86 ± 0.230
$0.010 \pm 3.53 \times 10^{-4}$	137.77 ± 0.344
$0.014 \pm 3.60 \times 10^{-4}$	150.01 ± 0.374
$0.024 \pm 4.24 \times 10^{-4}$	162.30 ± 0.405
$0.038 \pm 5.56 \times 10^{-4}$	174.58 ± 0.436
$0.052 \pm 7.31 \times 10^{-4}$	183.73 ± 0.458
$0.071 \pm 9.43 \times 10^{-4}$	193.02 ± 0.482
$0.084 \pm 1.10 \times 10^{-3}$	199.25 ± 0.497
$0.103 \pm 1.33 \times 10^{-3}$	205.41 ± 0.512
$0.126 \pm 1.62 \times 10^{-3}$	211.62 ± 0.528
$0.146 \pm 1.89 \times 10^{-3}$	217.33 ± 0.542
$0.182 \pm 2.34 \times 10^{-3}$	223.50 ± 0.558
$0.226 \pm 3.09 \times 10^{-3}$	229.72 ± 0.573

Table IV
Stress and Strain for Long Wire

Strain (mm/mm)	Stress (MPa)
$9.78 \times 10^{-4} \pm 1.03 \times 10^{-4}$	0.00 ± 0.016
$1.43 \times 10^{-3} \pm 1.10 \times 10^{-4}$	30.61 ± 0.078
$1.60 \times 10^{-3} \pm 1.07 \times 10^{-4}$	42.89 ± 0.108
$1.79 \times 10^{-3} \pm 1.05 \times 10^{-4}$	55.01 ± 0.138
$1.97 \times 10^{-3} \pm 1.12 \times 10^{-4}$	66.51 ± 0.167
$2.19 \times 10^{-3} \pm 1.32 \times 10^{-4}$	78.65 ± 0.196
$2.57 \times 10^{-3} \pm 2.39 \times 10^{-4}$	90.70 ± 0.227
$2.89 \times 10^{-3} \pm 2.45 \times 10^{-4}$	102.16 ± 0.255
$3.85 \times 10^{-3} \pm 1.56 \times 10^{-4}$	114.17 ± 0.285

Table IV Continued
Stress and Strain for Long Wire

Strain (mm/mm)	Stress (MPa)
$4.40 \times 10^{-3} \pm 2.37 \times 10^{-4}$	120.31 ± 0.300
$6.28 \times 10^{-3} \pm 1.85 \times 10^{-4}$	120.31 ± 0.315
$9.02 \times 10^{-3} \pm 2.49 \times 10^{-4}$	131.89 ± 0.329
$0.013 \pm 2.33 \times 10^{-4}$	137.64 ± 0.344

The above results are raw data points from the experiment that did not account for the stress and strain caused by the hanger. The relationship between stress and strain of 32 AWG copper wire from the data above is shown below in Fig. 4 for both wires.

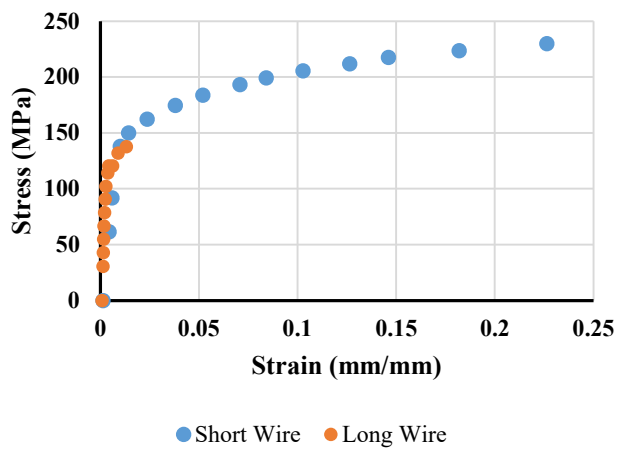

Fig. 4. Stress versus strain for short and lone wire.

The elastic region of the long wire was graphed separately in Fig. 5 below to determine the 0.002 offset yield point and modulus of elasticity.

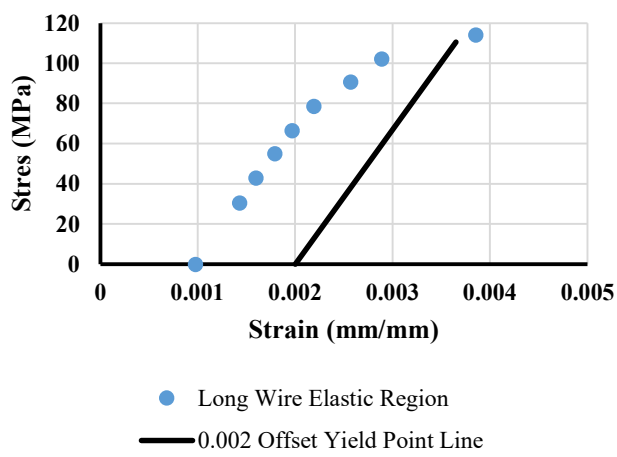

Fig. 5. Elastic region of long wire and 0.002 offset yield point line for stress versus strain of copper.

The modulus of elasticity, E, yield stress, σ_y, and offset yield stress, σ_{oy}, for 32 AWG copper were found from an analysis of the above graph. The ultimate tensile stress, σ_u, was found from the short wire stress-strain curve in Fig. 4. The results for modulus of elasticity, yield stress, and ultimate stress are shown in Table V.

Table V
Modulus of Elasticity, Yield Stress and Ultimate Stress of 32 AWG Copper Wire

Parameter	Value	Units
E	66.16 ± 17.94	GPa
σ_y	102.16	MPa
σ_{oy}	110.49	MPa
σ_u	229.72 ± 0.573	MPa

The given parameters used by LabVIEW to calculate all the above results are detailed in Table VI below. These parameters include the cross-sectional area of the copper wire, A, the LVDT sensitivity, S, the calibration constant, C, the initial lengths of both wires, L_{short} and L_{long}, and the tared LVDT voltages for both wires, $V_{LVDT,tare,short}$ and $V_{LVDT,tare,long}$. Also included is the weight of the hanger, W_h, which was relevant for obtaining a more accurate stress-strain curve as will be presented in the discussion. Finally, pi and the diameter of the wire, π and d, are included for a theoretical area uncertainty calculation despite LabVIEW not calculating the area based off these parameters.

Table VI
Given Parameters Used to Calculate Results

Parameter	Value	Units
A	$0.032 \pm 7.98 \times 10^{-5}$	mm^2
S	2.0619 ± 0.0015	V/mm
C	0.485 ± 0.0064	mm/V
L_{short}	41.37 ± 0.5	mm
L_{long}	109.92 ± 0.5	mm
$V_{LVDT,tare,short}$	4.768 ± 0.004	V
$V_{LVDT,tare,long}$	4.549 ± 0.004	V
W_h	0.248 ± 0.0005	N
π	3.14159 ± 0.000002	none
d	0.2019 ± 0.00025	mm

IV. Discussion

This experiment utilized load controlled tensile testing to analyze the stress-strain relationship of annealed 32 AWG copper wire. A hanger was connected to the bottom of the wire which was secured by a clamp and post at the top of the loading apparatus. The bottom of the hanger had a metal core that displaced inside an LVDT generating a differential voltage that was sent to the LabVIEW program through a DAQ to determine strain. Stress was calculated by the LabVIEW program based off the area and incremental weight inputs using (1). A short wire was first tested until it reached a breaking point to determine the ultimate strength of the copper wire. Then, a long wire was tested to obtain more accurate data on the elastic region and yield point of the copper wire.

The overall shape of the load versus displacement graph (Fig. 3) is very similar to that of the stress versus strain graph (Fig. 4). The reason behind this similarity is that load is equal to stress times a constant, A, or cross-sectional area of wire, and elongation is equal to strain times a constant, L, or initial length of the wire. Since both parameters, load and elongation, are

constant multiples of stress and strain, the graphed relationship follows the same shape. In a true stress analysis, the cross-sectional area of the wire would not be a constant value due to necking in the plastic region. However, only engineering stress was used in this experiment and thus the area was considered constant, allowing for the constant multiple relationship of load and stress.

A small strain value was present for the zero stress data points taken before loading weight on the hanger (Table III) (Table IV). Although no stress was being applied, this small strain was likely caused by slip in the wire or a taring issue that did not fully cancel out the strain from the hanger. Slight adjustments were made to try to reduce strain at this point, but it proved difficult to completely remove. Another issue with the raw stress and strain data is that it did not account for the stress and strain caused by the hanger weight. However, these values were added to the data afterward using the following method to generate a more accurate stress-strain curve.

Before the hanger weight adjustments could be made, an analysis of the long wire elastic region (Fig. 5) was necessary to determine the modulus of elasticity. By observation, the most linear portion of the long wire elastic region were between the second and fifth data points in Table IV. Therefore, the slope was taken between the fifth point and the second point to determine the modulus of elasticity. The same four stress and strain data points used to solve the modulus of elasticity were also used to determine its uncertainty in a Monte Carlo simulation [4]. The stress caused by the hanger weight was calculated by dividing its weight by the cross-sectional area of the wire. Then, the strain caused by the hanger weight was calculated by dividing that stress by the previously calculated modulus of elasticity. These stress and strain values caused by the hanger weight were added to every stress and strain value from the raw data to produce the more accurate stress strain graph shown in Fig. 6 below.

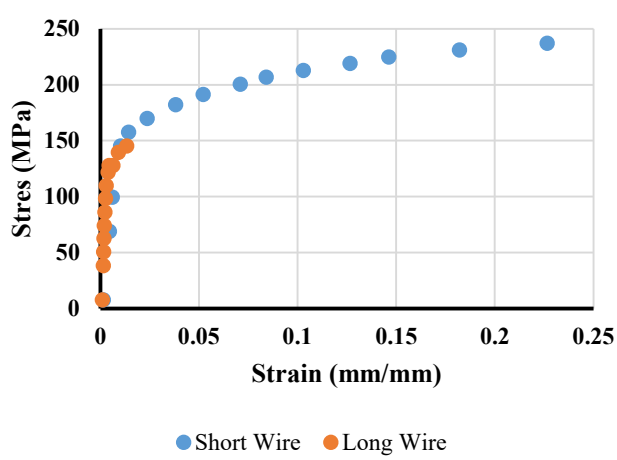

Fig. 6. Stress and strain curve of copper wire adjusted for hanger weight.

In comparison to the raw data stress-strain curve (Fig. 4), the graph above is slightly shifted up and to the right due to the addition of the hanger induced stress and strain to each point. Since the same values were added to every point, no changes in the overall shape of the curve occurred.

While the long wire elastic region graph (Fig. 5) was used to find the modulus of elasticity, it also functioned as a means to determining a yield point and offset yield point for the copper wire. The yield point from the raw data was determined by observing where the shape of the graph drastically changed. After the eighth long wire data point, an abnormal horizontal gap appeared before the ninth data point (Fig. 5). This gap was caused by a change in behavior as strain began to increase at an accelerated rate compared to stress. This signified the transition from the elastic to plastic region where permanent deformation began. Therefore, the stress at the eighth data point equal to 102.16 MPa was approximated as the yield stress of the copper wire. Next, the offset yield point was estimated by plotting a line beginning at 0.002 strain and zero stress with a slope equal to the previously calculated modulus of elasticity. The point at which this line intersected the trajectory of the long wire stress-strain curve was approximated as the offset yield point and was equal to 110.49 MPa. The offset yield point provides an overestimation of the actual yield point to serve as a conservative approximation to reduce the chance of materials yielding in analytical applications.

The ultimate tensile stress of the copper wire was found by observing the highest point on the short wire stress-strain curve, which happened to be the last point with a stress equal to 229.72 ± 0.573 MPa. Although roughly similar, the material properties calculated in this experiment were still considerably off from the generally accepted values for the copper wire used which are a modulus of elasticity, yield stress, and ultimate tensile stress equal to 110 GPa, 33.3 MPa and 210 MPa, respectively [2]. The reason for this disparity lies in the limitations of this experiment and the multiple sources of uncertainty.

There were multiple limitations to this experiment that affected the accuracy of results. First off, the measurement of the initial wire length was found by measuring the center of the two posts securing the wire with a digital caliper. Although the digital caliper was accurate, there was no method of accurately finding the center of the posts. Therefore, the center of the posts was visually estimated leading to an increase in caliper uncertainty. This uncertainty contributed to an increased uncertainty in every strain calculation that followed due to the parameter's dependence on initial length.

The next limitation involved the resolution of the ADC with recording smaller voltage values as the LVDT approached the end of its range. An observation of Tables I and II reveal an increase in uncertainty for the low LVDT voltage measurements. The sample window was set to ±5.12 volts to contain the ±5 V range of the LVDT. The resolution of this sample window was not optimal for the smaller voltage values, however, it could not be decreased or else saturation would occur for the higher values.

Another limitation was the creeping that occurred after the ultimate stress that made it extremely difficult to get accurate data near the end of the small wire stress-strain curve. The creep led the wire to gradually deform at a constant stress value. Therefore, the wire would break before an additional weight could be added to provide more accurate data for the end behavior of the curve. In addition to creep, slip of the wire sometimes still occurred despite a proper setup and proofing

load application. As a result, some data points appeared shifted or misplaced, or completely new data needed to be taken in extreme cases.

The large surface area to volume ratio of the copper wire was another contributing factor to the disparity between the calculated and actual material properties. The reason for this is that any small imperfection in the surface of the wire could contribute to a significant change in strength at that location. Certain surface defects may have led to a greater increase in strength during work hardening, while other defects may have reduced the amount of stress the wire could withstand leading to a lower yield and ultimate stress.

Multiple changes could be made to this experiment to increase the accuracy of the results. First off, the posts on the loading apparatus could be made with a divot at their centers. This way the calipers would be able to more accurately measure the initial wire lengths by securing their tips into the divots. Next, a better apparatus could be used to secure the wire and prevent it from slipping. For example, a clamp that can be screw tightened at the top may prevent slip better than the hand tightened clamp used in this experiment. Lastly, the temperature of the test area could be reduced to minimize creep as this phenomenon accelerates at higher temperatures.

V. Conclusion

This experiment successfully generated a stress-strain relationship for annealed 32 AWG copper wire. A short wire was first tested to determine the ultimate tensile strength of the copper wire which was equal to 229 ± 0.573 MPa. A long wire was then tested to analyze the elastic region and yield point of the wire generating the results of a yield stress equal to 102.16 MPa, an offset yield stress equal to 110.49 MPa and a modulus of elasticity equal to 229.72 ± 0.573 GPa. While the experiment provided reasonable estimates of the material properties of the copper wire, there were several factors affecting the accuracy of the results. Limitations such as wire slip, creep, ADC resolution, and human measurement errors made it difficult to produce consistent and accurate data. Additionally, multiple sources of uncertainties—given, statistically determined, RSS propagated, and Monte Carlo simulated—contributed to the lower accuracy and precision of results.

Appendix

All parameters used in this experiment had an associated uncertainty. Four different methods of determining uncertainty were utilized depending on how the parameter was measured or calculated. First, some uncertainties were provided by manufacturers or known by rule of thumb (Table VII). Statistical uncertainties with a 95% confidence interval were found for the voltage measurements from the DAQ based off 1000 samples (Table VIII). The uncertainty for the LVDT voltage is equal to the average uncertainty of the differential voltage associated with each data point. The uncertainties for parameters dependent on other variables were found using RSS propagation equations that considered all contributing uncertainties (Table IX). Finally, a Monte Carlo simulation was used to calculate the uncertainty of the modulus of elasticity. The uncertainties of the four stress and strain values used in the simulation as well as the uncertainty of the modulus of elasticity are detailed in Table X.

Table VII
Given Uncertainties

Parameter (units)	μ	Uncertainty	Source
S (V/mm)	μ_S	± 0.0015	Manufacturer [5]
d (mm)	μ_d	± 0.00025	Manufacturer [3]
π (none)	μ_π	± 0.000002	Rule of Thumb [3]
P (N)	μ_P	± 0.0005	Manufacturer [4]
l_{short} (mm)	μ_{lshort}	± 0.5	Rule of Thumb [3]
l_{long} (mm)	μ_{llong}	± 0.5	Rule of Thumb [3]

Table VIII
Statistical Uncertainties

Parameter (units)	μ	Uncertainty	# Samples
V_{LVDT} (V)	μ_{VLVDT}	± 0.008	1000
$V_{LVDT,tare,short}$ (V)	$\mu_{VLVDT,tare,short}$	± 0.004	1000
$V_{LVDT,tare,long}$(V)	$\mu_{VLVDT,tare,long}$	± 0.004	1000

Table IX
RSS Propagated Uncertainties

Parameter	μ	Uncertainty	Units
C	μ_C	± 0.0064	(mm/V)
A	μ_A	± 7.98 x 10^{-5}	mm^2
ΔL	$\mu_{\Delta L}$	± 0.021	mm
σ_x	$\mu_{\sigma x}$	± 0.331	MPa
ε_x	$\mu_{\varepsilon x}$	± 6.52 x 10^{-4}	(mm/mm)

The uncertainty for every stress and strain point was calculated with the respective RSS propagation equations separately. Each of these individual uncertainties can be observed in Tables III and IV for the short wire and long wire, respectively. The uncertainty for axial stress and strain displayed in the table above is an average of all the separate uncertainties as the values varied. This same method was also applied to the elongation, ΔL, uncertainty reporting.

Table X
Monte Carlo Simulation Uncertainties

Parameter	μ	Value and Uncertainty	Units
ε_1	$\mu_{\varepsilon 1}$	1.43 x 10^{-3} ± 1.10 x 10^{-4}	(mm/mm)
ε_2	$\mu_{\varepsilon 2}$	1.60 x 10^{-3} ±1.07 x 10^{-4}	(mm/mm)
ε_3	$\mu_{\varepsilon 3}$	1.79 x 10^{-3} ± 1.05 x 10^{-4}	(mm/mm)
ε_4	$\mu_{\varepsilon 4}$	1.97 x 10^{-3} ± 1.12 x 10^{-4}	(mm/mm)
σ_1	$\mu_{\sigma 1}$	30.61 ± 0.078	MPa
σ_2	$\mu_{\sigma 2}$	42.89 ± 0.108	MPa

Table X Continued
Monte Carlo Simulation Uncertainties

Parameter	μ	Value and Uncertainty	Units
σ_3	$\mu_{\sigma 3}$	55.01 ± 0.138	MPa
σ_4	$\mu_{\sigma 4}$	66.51 ± 0.167	MPa
E	μ_E	66.16 ± 17.94	GPa

The eight stress and strain values and their uncertainties were input into an excel file that performed a Monte Carlo simulation with thousands of samples [4]. This simulation generated the standard deviation of the modulus of elasticity, *E,* to which a 95% confidence interval was applied to determine the uncertainty reported above.

RSS Propagation Uncertainty Equations:

Calibration Constant Uncertainty, μ_C

$$\mu_C = \frac{\partial \mathrm{C}}{\partial \mathrm{S}} \mu_S \tag{7}$$

Elongation (ΔL) Uncertainty, $\mu_{\Delta L}$

$$\mu_{\Delta L} = [\frac{\partial \Delta L^2}{\partial V_{LVDT,tare}} \mu_{LVDT,tare}^2 + \frac{\partial \Delta L^2}{\partial V_{LVDT}} \mu_{LVDT}^2 + \frac{\partial \Delta L^2}{\partial V_S} \mu_S^2]^{1/2} \tag{8}$$

Strain Uncertainty, $\mu_{\varepsilon x}$

$$\mu_{\varepsilon x} = [\frac{\partial \varepsilon_x^2}{\partial \Delta \mathrm{L}} \mu_{\Delta L}^2 + \frac{\partial \varepsilon_x^2}{\partial \mathrm{L}} \mu_L^2]^{1/2} \tag{9}$$

Stress Uncertainty, μ_σ

$$\mu_{\sigma x} = [\frac{\partial \sigma_x^2}{\partial \mathrm{P}} \mu_P^2 + \frac{\partial \sigma_x^2}{\partial \mathrm{A}} \mu_A^2]^{1/2} \tag{10}$$

Area Uncertainty, μ_A

$$\mu_A = [\frac{\partial A^2}{\partial \mathrm{d}} \mu_d^2 + \frac{\partial A^2}{\partial \pi} \mu_\pi^2]^{1/2} \tag{11}$$

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